

Quantum Biomarker Algorithms for Multimodal Cancer Data
Phase 2 Milestone 1 Report (December 20, 2024)
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1 Executive Summary and Recap

Our ongoing Phase 2 research has yielded promising results, demonstrating substantial efficiency gains and accuracy improvements to cancer classification tasks, as we pursue our ultimate goal of demonstrating quantum advantage. Through our novel Hyper-RQAOA approach – which represents the first ever demonstration of parameter transfer for RQAOA – we achieved a 1000X reduction in computational resources while delivering superior energy solutions compared to baseline RQAOA methods. In cancer diagnostics, our feature selection algorithms improved tissue-of-origin prediction accuracy by 25% across 30 cancer types when selecting 6 features from the entire **Pancan** dataset. These results, in addition to recent experimental progress demonstrating lightweight error detection on logical qubits which resulted in 15X error reduction on random circuits and 12X error reduction for ground state preparation [1], gives us confidence that we can demonstrate quantum advantage in Phase 3. With Inflektion’s multi-platform Superstaq software [2], we will target demonstrations across multiple modalities, including our own neutral atom platform, Quantinuum, and IBM. This report provides a snapshot of our ongoing Phase 2 research and describes our plans for final Phase 2 experiments and future Phase 3 demonstrations, including:

PCBO-Tournament Evaluation and Improvement: We evaluated our hybrid feature selection algorithm on the **Pancan** data set, achieving a 25% improvement over existing methods when selecting 6 features from a pool of 23,225 features for 30-cancer tissue of origin classification (Section 2). We also focused on optimizing the PCBO cost functions to better leverage higher-order information within the data, evaluating our implementations for smaller feature sets (Section 2.1) and theoretical models to predict the presence of valuable higher-order information in multimodal cancer data sets (Section 2.2).

Classical, Quantum Annealing, and Hybrid Solvers: In preparation for future quantum advantage demonstrations, we conducted a survey of classical state-of-the-art solvers for PCBO problems (Section 3.1). Additionally, we also tested our PCBO problems on D-Wave’s quantum annealer (Section 3.2), where we found that while the all-to-all connectivity of our problems posed challenges for direct implementation, the D-Wave hybrid solver showed promising results.

Resource Optimization: A primary focus of our initial Phase 2 research was the pursuit of resource efficiency gains through parameter transfer techniques. We conducted thorough studies of parameter transfer for our PCBO problems using QAOA (Section 4.2), Lateral-RQAOA (Section 4.3.1), and the novel Hyper-RQAOA algorithm (Section 4.3.2). **Our work represents the first studies of parameter concentration in the RQAOA.** Furthermore, while prior work has primarily considered MaxCut problems on random instances drawn from an abstract problem class, our work focuses on concrete PCBO problem instances constructed from real-world multimodal cancer data sets. Overall, we found that **parameter transfer can effectively initialize larger RQAOA instances using smaller classical simulations – resulting in lower energy solutions using 1000X fewer resources.** The resource savings enabled by parameter transfer imply that our hybrid feature selection algorithm may be evaluated on quantum hardware much sooner than otherwise possible. This has important implications for our planned Phase 2 simulations, future Phase 3 hardware experiments, and the potential for applying our hybrid feature selection algorithm to cancer data sets in a real-world setting.

Understanding the Potential for Quantum Advantage on Cancer Data Sets: We also report on progress developing classical algorithms approximating "strong k-contextuality" in classical data sets. This metric helps identify data sets that could benefit most from quantum machine learning models by determining the minimum memory requirements for accurate classical modeling. We successfully tested these approximation algorithms on both GHZ datasets with known solutions and small DNA-mRNA samples from the **Pancan** dataset (Section 5).

Ongoing and Future Work: Effective parameter transfer has particularly important implications for our future work, supporting the feasibility of near-term hardware demonstrations of our hybrid feature selection algorithm. These results pave the way for practical applications of quantum computing in cancer research and establish a pathway to viable hardware demonstrations in Phase 3 by leveraging recent progress in error mitigation, detection, and correction (Section 6).



Figure 1: Overview of our hybrid data processing pipeline. We implemented the entire pipeline end-to-end in order to adequately evaluate our PCBO-Tournament feature selection algorithm.

1.1 Recap of the Hybrid Data Processing Pipeline

1.1.1 Data Preprocessing

In Phase 1 we developed novel preprocessing techniques for the mRNA and pathomics data modalities used in our data sets. The raw mRNA features from The Cancer Genome Atlas (TCGA) [3] are positive real values, and we discretize these features into five bins with integer values 0–4 via quantile discretization. The raw pathomics data come in the form of gigapixel slide images, we discretize and standardize the features across patients using a combination of feature extraction via deep neural networks (based on the `slideflow` tools developed in Professor Pearson’s lab [4]) and quartile discretization.

1.1.2 Feature Selection

In Phase 1 we shifted our primary focus to developing hybrid algorithms for feature selection because our research indicated that this particular application has a higher potential for achieving quantum advantage using the quantum hardware expected to come online within the next 5 years. The feature selection task, over F total features, can be expressed mathematically as a Polynomial Constrained Binary Optimization (PCBO) problem of the form:

$$\arg \min_{\mathbf{x}} \left(\sum_i^F C(i) x_i + \sum_{i < j}^F C(i, j) x_i x_j + \sum_{i < j < k}^F C(i, j, k) x_i x_j x_k + \dots \right) \quad (1)$$

subject to the constraints: $|\mathbf{x}|_1 = N \in \mathbb{Z}, \quad \mathbf{x} \in \mathbb{Z}_2^N$

Here, the aim is to “turn on” N bits of the binary vector $\mathbf{x}$, and the indices of the $x_i = 1$ bits correspond to the features f_i in the selected feature set. The functions $C(\cdot)$ yield the coefficients defining the optimization problem. We use the mutual information between features and class labels, as well as between pairs and triplets of features, to define the coefficient values for the feature selection problem. We developed the PCBO-Tournament algorithm as a hybrid approach to feature selection on multimodal cancer data sets. We use the Recursive Quantum Approximate Optimization Algorithm (RQAOA) [5] as the key hybrid computational kernel within the algorithm to find low energy solutions to PCBO problems constructed over the data features. However, because qubits are a scarce resource and real-world cancer data sets often contain many thousands of features we also introduced a “tournament” aspect to the algorithm which operates by taking the original, full data set and generating many smaller subproblems that can be solved with near-term quantum hardware (or via classical simulation).

1.1.3 Machine Learning

In our current pipeline, Step 3 of the data pipeline is composed of simple classical models (logistic regression and support vector machines) which provides a fixed baseline classifier allowing us to assess the performance of the hybrid feature selection algorithms we are developing. However, in collaboration with Professor Chong’s graduate students at UChicago, we have also continued to explore the potential for quantum machine learning algorithms in this application space, and their potential to replace the classical models in Step 3. We describe some of these latest results in Section 5 below.

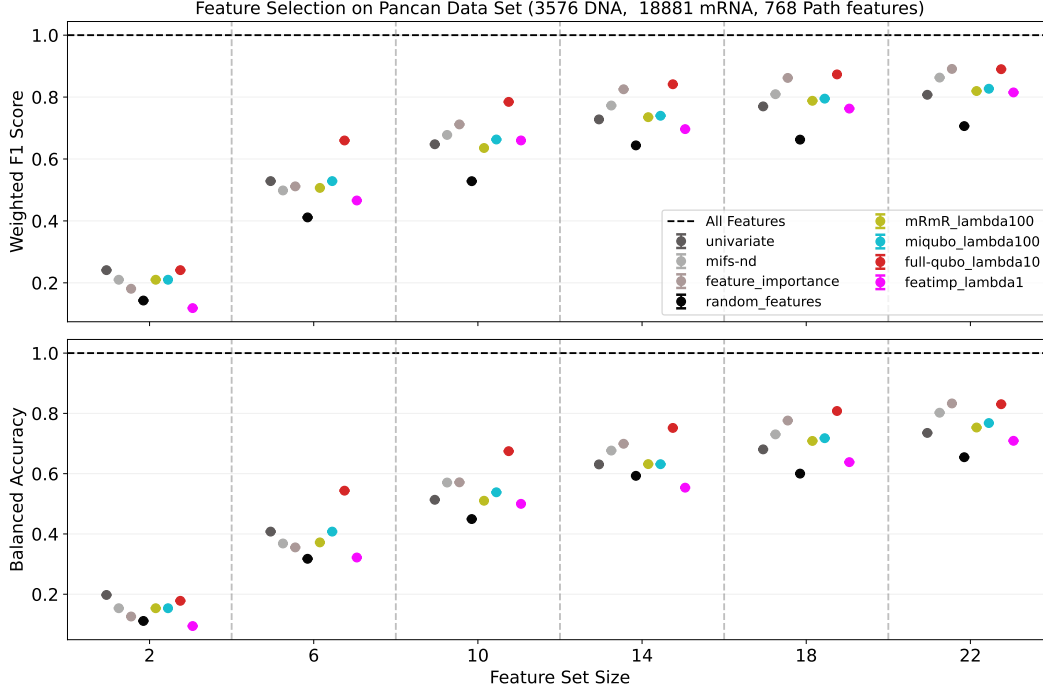


Figure 2: Comparison of performance amongst classical (gray and pink) and the PCBO-Tournament feature selection algorithms on the **Pancan** data set. The **full-qubo** variant of the PCBO-Tournament algorithm is able to select the highest performing feature sets of sizes 6, 10, 14, and 18. These results suggest that the PCBO-Tournament algorithm is best suited to identifying small and informative feature sets. For example, when selecting 6-feature sets, **full-qubo** achieves a 25% improvement in F1 score (0.660 versus 0.528 F1 scores) compared to the next best algorithm **univariate** [6].

2 PCBO-Tournament Evaluation

Data set name	Data Summary	Target learning problem
Pancan [3]	7,171 cancer patients; 3,576 binary DNA features; 18,881 integer mRNA features; 768 integer pathomics features	30-cancer tissue of origin classification

Table 1: Overview of the multimodal data set and learning task used thus far in Phase 2 evaluations.

We evaluated the PCBO-Tournament feature selection algorithm on the **Pancan** data set. This data set is a challenging extension to the **Pansquamous** data set evaluated in Phase 1. Where the **Pansquamous** data focused on classifying patients with one of three different squamous-cell carcinomas, the **Pancan** data set is a larger 30-class classification task across a range of cancer types from the TCGA data base. Furthermore, we extended the **Pancan** data set since Phase 1 by integrating pathomics features into the rows of data points. Due to the availability of pathomics data varying across patients in the TCGA data base, the total number of cancer patients in this data set was reduced by 842, but now each patient is associated with a unified set of features across genomics, transcriptomics, and pathomics modalities.

Figure 2 shows the results of our simulations which applied the PCBO-Tournament algorithm to the **Pancan** data set to select feature sets of size 2–22. We observed that the **full-qubo** formulation of the PCBO-Tournament was able to select the highest performing feature sets of size 6, 10, 14, and 18. Beyond these sizes, the PCBO-Tournament performs as well as the feature importance algorithm and both methods

approach 100% classification accuracy as the feature set size approaches 100.

Figure 2 also contains important information concerning the impact that the tournament structure has on the overall performance of the feature selection algorithm. Namely, we also evaluated a “**feature importance tournament**” algorithm (pink `featimp_lambda1` points in Fig 2) which uses the same subproblem generation structure as the PCBO-Tournament, but it uses the **feature importance** algorithm to decide which features move on to the next round. Compared to baseline **feature importance** (gray-brown points in Fig 2) which observes the entire data set and selects a feature set in a single pass, the **feature importance tournament** consistently produces lower accuracy feature sets.

This evaluation provides us with two key insights: first, it is interesting to see that the **full-qubo** instantiation of the PCBO-Tournament algorithm still produces high accuracy feature sets despite utilizing random subproblem generation in each round of the tournament. Second, these results indicate a high potential for improvement of the PCBO-Tournament algorithm by improving the tournament structure. This is a primary objective of our research in the second half of Phase 2. In the first half of Phase 2, we chose to focus on improving the formulation of the PCBO cost function itself in order to better utilize the higher-order information available in the data set. Intuitively, we wanted to ensure that we are solving each PCBO subproblem as well as possible, and then we may realize complimentary improvements by altering the tournament structure. The results of this research are presented in Sections 2.1 and 2.2.

2.1 Improving the PCBO Formulation to Better Leverage Higher-Order Information

In the first half of Phase 2 we conducted experiments and analyses focused on modifying the formulation of the PCBO cost function to yield more informative feature sets. We defined the following four PCBOs, taking advantage of second- and third-order information between variables in different ways.

Total PCBO (T-PCBO):

$$\begin{aligned} \varphi(\vec{x}) = & \sum_i -I(f_i, y) x_i + \sum_{i,j} (\lambda I(f_i, f_j) + \bar{\lambda} I(f_i, f_j | y) + \hat{\lambda} I(f_i, y | f_j)) x_i x_j \\ & + \sum_{i,j,k} (\mu I(f_i, f_j, f_k) + \bar{\mu} I(f_i, f_j, f_k | y) + \hat{\mu} I(f_i, f_j, y | f_k)) x_i x_j x_k \end{aligned} \quad (2)$$

Target-Unconditioned PCBO (TU-PCBO):

$$\begin{aligned} \varphi(\vec{x}) = & \sum_i -I(f_i, y) x_i + \sum_{i,j} (\lambda I(f_i, f_j) + \hat{\lambda} I(f_i, y | f_j)) x_i x_j \\ & + \sum_{i,j,k} (\mu I(f_i, f_j, f_k) + \hat{\mu} I(f_i, f_j, y | f_k)) x_i x_j x_k \end{aligned} \quad (3)$$

Target-Unconditioned QUBO (TU-QUBO):

$$\varphi(\vec{x}) = \sum_i -I(f_i, y) x_i + \sum_{i,j} (\lambda I(f_i, f_j) + \hat{\lambda} I(f_i, y | f_j)) x_i x_j \quad (4)$$

Target-Unconditioned, Depressed PCBO (TUD-PCBO):

$$\varphi(\vec{x}) = \sum_i -I(f_i, y) x_i + \sum_{i,j,k} (\mu I(f_i, f_j, f_k) + \hat{\mu} I(f_i, f_j, y | f_k)) x_i x_j x_k \quad (5)$$

T-PCBO is a natural, third-order generalization of miquibo and mRmR. TU-PCBO, while less natural of a generalization due to conditioning on the target variable, is included for the sake of exploratory analysis. Lastly, TU-QUBO and TUD-PCBO¹ are included in our analysis for the sake of understanding the marginal contributions of second- and third-order terms to the PCBO energy function for a given set of features.

¹A depressed polynomial of degree k is one whose $(k-1)$ th-order term is zero.

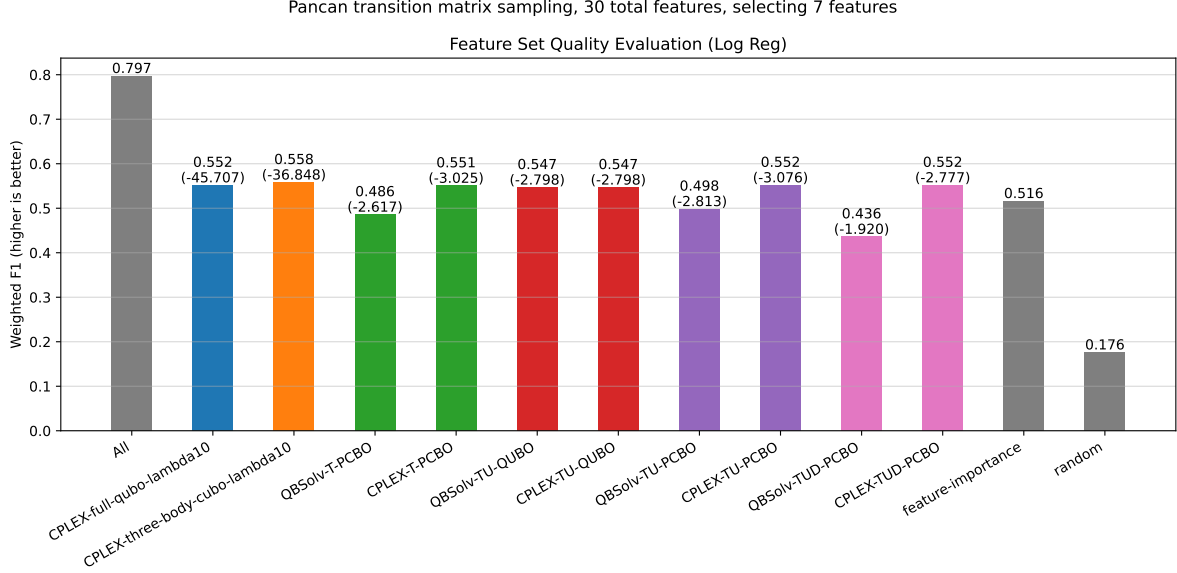


Figure 3: Evaluation of different PCBO cost function formulations on an example 30-feature problem drawn from the **Pancan** data set. The number above each bar denotes the weighted F1 score, and the negative number in parentheses denotes the energy that the solver attained (specific energies can only be compared between bars of the same color). Notably, the PCBOs containing third-order terms attain the highest accuracy (achieving a 7% improvement over **feature importance**), but only when the CPLEX solver is used. The consistent pattern of “finding a lower energy solution yields a higher accuracy feature set” motivates the development of better quantum solvers for this application.

To assess the ability of our new CUBO formalism to select small, informative feature subsets, we generated subsets of 30 features from **Pancan** using a structured, randomized approach (Sec. 2.1.1). For each of these sets of 30 features, we looked at all possible subsets of seven features, computing the balanced accuracy of predicting the target variable² using a three-to-one training-test split. For each of miqubo, mRmR, full QUBO, and the PCBOs from Section 2.1, the optimal coefficient weights $\lambda, \bar{\lambda}, \hat{\lambda}, \mu, \bar{\mu}, \hat{\mu}$ (where applicable) were fit to the balanced accuracy by linear regression. Keeping the coefficient weights fixed at these learned values, the minimum energy feature subset of size seven for each PCBO was found using solvers from the CPLEX library [7]. Figure 3 shows these results.

2.1.1 Small feature set generation by graph diffusion

Feature sets of size thirty were generated from the total set of **Pancan** features using the following strategy. We begin by fixing $a, b \geq 0$ and sampling thirty features X_i with probability proportional to $(I(X_i; Y))^a$, where Y is the target variable. We then, for all i , sample a feature Z_i with probability proportional to $(I(X_i; Z_i))^b$. The resulting Z_i comprise the small feature set.

2.2 Han inequality and ideal feature regimes

Say we have random variables $X_1, \dots, X_n$ and target random variable Y . Further, say we want to reorder the indices $1, \dots, n$ such that the mutual information between $(X_1, \dots, X_k)$ and Y is maximized. Observe that

$$I(X_1, \dots, X_k; Y) = H(X_1, \dots, X_k) + H(Y) - H(X_1, \dots, X_k, Y) \quad (6)$$

$$= H(X_1, \dots, X_k) + H(Y) - H(X_1, \dots, X_k | Y) - H(Y) \quad (7)$$

$$= H(X_1, \dots, X_k) - H(X_1, \dots, X_k | Y) \quad (8)$$

²The target variable for **Pancan** is tissue of origin. See the third column of Table 1.

In practice, we rarely compute the joint entropy of several variables directly, but instead compute mutual and interaction information between small subsets of features. The following three lines give forms for the joint entropy using first-, second-, and third-order entropies, respectively:

$$H(X_1, \dots, X_k) = \frac{1}{k} \sum_{i=1}^k \left(H(X_i) + H(X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_k \mid X_i) \right) \quad (9)$$

$$= \frac{1}{\binom{k}{2}} \sum_{\{i,j\} \subset [1,k]} \left(H(X_i, X_j) + H(\{X_1, \dots, X_k\} \setminus \{X_i, X_j\} \mid \{X_i, X_j\}) \right) \quad (10)$$

$$= \frac{1}{\binom{k}{3}} \sum_{\{i,j,l\} \subset [1,k]} \left(H(X_i, X_j, X_l) + H(\{X_1, \dots, X_k\} \setminus \{X_i, X_j, X_l\} \mid \{X_i, X_j, X_l\}) \right) \quad (11)$$

Note that any convex combination of the above righthand sides also provides a formula for $H(X_1, \dots, X_k)$. Using all of this, how do we find problems for which maximizing $\mathcal{I} := I(X_1, \dots, X_k; Y)$ using first-, second-, and third-order information yields better performance than maximizing using first- and second-order information alone? First, note that the maximum value of $\mathcal{I}$ is proportional to its mean value. Defining $\tilde{X}_i$ as the variable X_i conditioned³ on Y , and letting $\alpha_1, \alpha_2, \alpha_3 > 0$ satisfy $\alpha_1 + \alpha_2 + \alpha_3 = 1$, we get the following.

$$\begin{aligned} \bar{\mathcal{I}} &= \frac{1}{\binom{n}{k}} \sum_{\{i_1, \dots, i_k\} \subset [1,n]} \left(H(X_{i_1}, \dots, X_{i_k}) - H(\tilde{X}_{i_1}, \dots, \tilde{X}_{i_k}) \right) \\ &= \frac{1}{\binom{n}{k}} \sum_{\{i_1, \dots, i_k\} \subset [1,n]} \left(\frac{\alpha_1}{k} \sum_{j=1}^k \left(H(X_{i_j}) + H(X_{i_1}, \dots, X_{i_{j-1}}, X_{i_{j+1}}, \dots, X_{i_k} \mid X_{i_j}) \right. \right. \\ &\quad \left. \left. - H(\tilde{X}_{i_j}) - H(\tilde{X}_{i_1}, \dots, \tilde{X}_{i_{j-1}}, \tilde{X}_{i_{j+1}}, \dots, \tilde{X}_{i_k} \mid \tilde{X}_{i_j}) \right) \right. \\ &\quad \left. + \frac{\alpha_2}{\binom{k}{2}} \sum_{\{j,l\} \subset [1,k]} \left(H(X_{i_j}, X_{i_l}) + H(\{X_{i_1}, \dots, X_{i_k}\} \setminus \{X_{i_j}, X_{i_l}\} \mid \{X_{i_j}, X_{i_l}\}) \right. \right. \\ &\quad \left. \left. - H(\tilde{X}_{i_j}, \tilde{X}_{i_l}) - H(\{\tilde{X}_{i_1}, \dots, \tilde{X}_{i_k}\} \setminus \{\tilde{X}_{i_j}, \tilde{X}_{i_l}\} \mid \{\tilde{X}_{i_j}, \tilde{X}_{i_l}\}) \right) \right. \\ &\quad \left. + \frac{\alpha_3}{\binom{k}{3}} \sum_{\{j,l,m\} \subset [1,k]} \left(H(X_{i_j}, X_{i_l}, X_{i_m}) + H(\{X_{i_1}, \dots, X_{i_k}\} \setminus \{X_{i_j}, X_{i_l}, X_{i_m}\} \mid \{X_{i_j}, X_{i_l}, X_{i_m}\}) \right. \right. \\ &\quad \left. \left. - H(\tilde{X}_{i_j}, \tilde{X}_{i_l}, \tilde{X}_{i_m}) - H(\{\tilde{X}_{i_1}, \dots, \tilde{X}_{i_k}\} \setminus \{\tilde{X}_{i_j}, \tilde{X}_{i_l}, \tilde{X}_{i_m}\} \mid \{\tilde{X}_{i_j}, \tilde{X}_{i_l}, \tilde{X}_{i_m}\}) \right) \right) \end{aligned}$$

Due to combinatorial considerations, the many-feature conditional entropy terms in the above summations are not computed in practice. Dropping these terms, we get the following approximation $\tilde{\mathcal{I}}$ for $\bar{\mathcal{I}}$.

$$\begin{aligned} \tilde{\mathcal{I}} &= \frac{1}{\binom{n}{k}} \sum_{\{i_1, \dots, i_k\} \subset [1,n]} \left(\frac{\alpha_1}{k} \sum_{i=1}^k \left(H(X_{i_i}) - H(\tilde{X}_{i_i}) \right) \right. \\ &\quad \left. + \frac{\alpha_2}{\binom{k}{2}} \sum_{\{i,j\} \subset [1,k]} \left(H(X_i, X_j) - H(\tilde{X}_i, \tilde{X}_j) \right) \right. \\ &\quad \left. + \frac{\alpha_3}{\binom{k}{3}} \sum_{\{i,j,l\} \subset [1,k]} \left(H(X_i, X_j, X_l) - H(\tilde{X}_i, \tilde{X}_j, \tilde{X}_l) \right) \right). \end{aligned}$$

³While conditioning one random variable on another cannot always induce a new random variable, it is well defined by interpreting the discrete target variable as a map $Y : \Omega \rightarrow \Sigma$, where Ω is the sample space for the measure space of the random variables $X_1, \dots, X_n, Y$ and Σ is the uniform measure space on the possible values of Y ; pairing Ω with the measure $Y^{-1}(\Sigma)$ induces a probability space with respect to which $\tilde{X}_1, \dots, \tilde{X}_n$ are random variables.

We can glean insight as to how to choose $\alpha_1, \alpha_2, \alpha_3$ using a theorem from Han [8]. This theorem states that the averages

$$h_i^{(n)} := \frac{1}{\binom{n}{i}} \sum_{\{i_1, \dots, i_k\} \subset [1, n]} \frac{H(X_{i_1}, \dots, X_{i_k})}{k}, \quad \tilde{h}_i^{(n)} := \frac{1}{\binom{n}{i}} \sum_{\{i_1, \dots, i_k\} \subset [1, n]} \frac{H(\tilde{X}_{i_1}, \dots, \tilde{X}_{i_k})}{k}$$

necessarily obey the relationship

$$h_1^{(n)} \geq \dots \geq h_n^{(n)}, \quad \tilde{h}_1^{(n)} \geq \dots \geq \tilde{h}_n^{(n)}. \quad (12)$$

We can concisely write

$$\tilde{\mathcal{I}} = \alpha_1 \left(h_1^{(n)} - \tilde{h}_1^{(n)} \right) + 2\alpha_2 \left(h_2^{(n)} - \tilde{h}_2^{(n)} \right) + 3\alpha_3 \left(h_3^{(n)} - \tilde{h}_3^{(n)} \right).$$

Given the above, when is it advantageous to use third-order information in addition to second-order information in a PCBO? We look at the case where α_1 is fixed; there exists a parameter α such that $\alpha_3 = \alpha$ and $\alpha_2 = (1 - \alpha)$. Differentiating with respect to α is given by the relationship

$$\frac{\partial \tilde{\mathcal{I}}}{\partial \alpha} = \left(3h_3^{(n)} - 3\tilde{h}_3^{(n)} \right) - \left(2h_2^{(n)} - 2\tilde{h}_2^{(n)} \right).$$

We can make this even more concise by defining $\Delta_3 := \left(h_3^{(n)} - \tilde{h}_3^{(n)} \right)$ and $\Delta_2 := \left(h_2^{(n)} - \tilde{h}_2^{(n)} \right)$:

$$\frac{\partial \tilde{\mathcal{I}}}{\partial \alpha} = 3\Delta_3 - 2\Delta_2. \quad (13)$$

Together, Equation (13) and the inequalities (12) give us a heuristic for saying whether leveraging third-order information in a PCBO approach yields better performance as compared to mere second-order information: namely, we assume leveraging third-order information is advantageous when $\Delta_3 > \frac{2\Delta_2}{3}$. The ranges of permissible values for $h_1^{(n)}, \dots, h_4^{(n)}, \tilde{h}_1^{(n)}, \dots, \tilde{h}_4^{(n)}$ are shown in Figure 4, and our heuristic for applying third-order information is visualized in Figure 5.

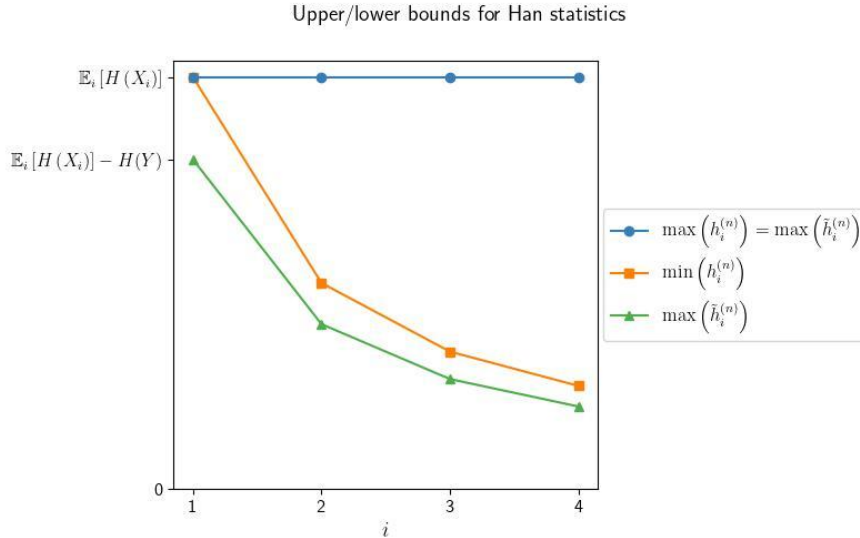


Figure 4: The possible ranges of values for $h_i^{(n)}, \tilde{h}_i^{(n)}$. These ranges constrain the regime wherein our heuristic states third-order information yields advantage over first- and second-order information alone (Fig. 5).

It is advantageous when $h_3^{(n)} - h_2^{(n)}$ is small and $\tilde{h}_3^{(n)} - \tilde{h}_2^{(n)}$ is large, where $h_i^{(n)}$ is the statistic from the Han Theorem for $X_1, \dots, X_n$ and $\tilde{h}_i^{(n)}$ is the analogous statistic for $\tilde{X}_1, \dots, \tilde{X}_n$.

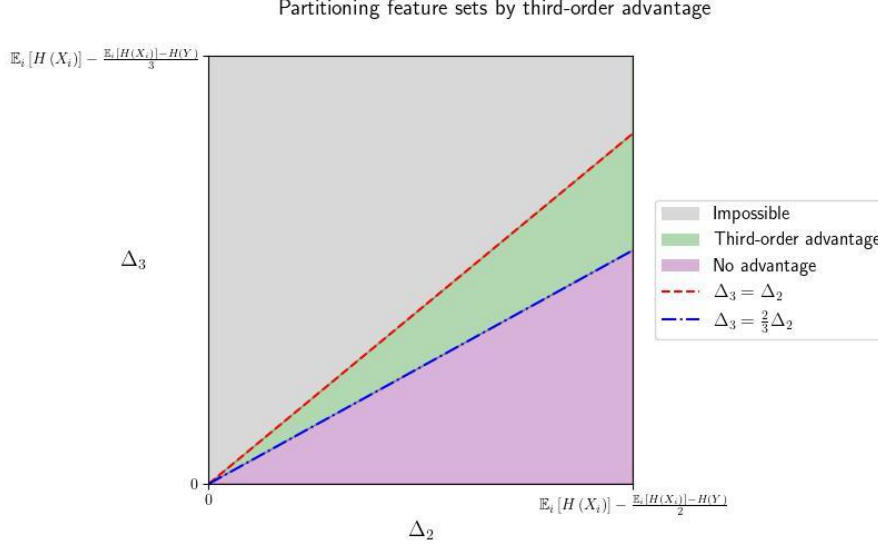


Figure 5: By reasoning of lower-order approximation of $\tilde{\mathcal{I}}$, there exist values of Δ_2, Δ_3 for which leveraging third-order information yields advantage over first- and second-order information alone.

3 Solving PCBOs with State-of-the-art Heuristics and Quantum Annealers

PCBO problems represent a fundamental and challenging class of optimization problems. In our research, we focus on PCBOs containing up to third-order terms, which we convert into unconstrained problems by introducing appropriate penalty terms. This transformation leads us to Cubic Unconstrained Binary Optimization (CUBO) problems. Solving CUBOs poses significant challenges for modern solvers across all hardware architectures – hence our interest in developing hybrid algorithms for this problem class. In collaboration with Professor Chong’s lab at the University of Chicago we have started to survey the current state-of-the-art classical *and* hybrid quantum-classical solvers for these problems, in preparation for future head-to-head comparisons with quantum hardware evaluations in Phase 3. The following sections summarize our exploration of state-of-the-art classical methods, for eventual comparison with quantum hardware experiments, and examines potential quantum approaches using the D-Wave quantum annealers.

3.1 Classical Solvers

Oftentimes classical algorithms tackle PCBO problems by reducing them to an equivalent version containing only quadratic terms. In fact, this is how the QBSolv [9] and CPLEX [7] solvers operate. The reduction of CUBO problems to quadratic models is achieved through quadratization [10]. The Rosenberg quadratization approach [11] provides a systematic method for this reduction.

Given a polynomial optimization problem

$$\min(P(\mathbf{x}) : \mathbf{x} \in \{0, 1\}^n), \quad (14)$$

a quadratized polynomial $Q(\mathbf{x}, \mathbf{y})$ where $\mathbf{y} \in \{0, 1\}^m$ maintains the same optimal value as P . For a term $cx_1x_2 \dots x_d$ in $P(\mathbf{x})$, where $c \in \mathbb{R}$, the reduction transforms it to $cx_1x_2 \dots x_{d-2}y$, with y constrained by $y = x_{d-1}x_d$. This constraint is encoded through the penalty term:

$$T = M(x_{d-1}x_d - 2x_{d-1}y - 2x_dy + 3y) \quad (15)$$

where $M \in \mathbb{R}$ represents the penalty weight. The efficiency of the resulting quadratic form depends on minimizing the number of terms affected by the quadratization procedure. While higher penalty weights reduce constraint violations, their optimal values depend on the solver characteristics. The resulting polynomial Q

is not unique, and minimizing the number of quadratized terms typically improves solver performance by reducing potential constraint violations.

Classical optimization methodologies encompass two primary categories: exact solvers and heuristics [12, 13, 14]. Exact solvers guarantee optimality or bounded suboptimality, while heuristics provide efficient solutions without optimality guarantees.

Exact Solvers – Branch and bound algorithms systematically explore the discrete search space by constructing cost bounds for tree branches, enabling efficient pruning when bounded costs exceed the best-found solution. Gurobi’s Integer Quadratic Programming solver [15] has demonstrated superior performance in comparisons with quantum approaches. For higher-order problems, quadratized PUBO with constraints in Gurobi or Mixed Integer Nonlinear Programming Solvers like IPOPT [16] serve as effective benchmarks.

Heuristics – Parallel Tempering with Isoenergetic Cluster Moves (PT+ICM) represents a leading classical heuristic [17]. The algorithm maintains multiple state replicates at varying temperatures, updated via the Metropolis rule:

$$\mathbb{P}(\text{accept } x_{\text{new}} | x_{\text{new}}, x_{\text{old}}) = \min[1, e^{\beta(x_{\text{new}} - x_{\text{old}})}] \quad (16)$$

where β denotes the inverse temperature. The method incorporates Houdayer Clusters for rejection-free non-local exploration [18]. For optimal performance, maintaining the CUBO formulation is preferred over quadratization, as constraint satisfaction relies on penalty terms or specialized update rules.

These algorithms represent the current classical state-of-the-art for PCBO optimization, and our ongoing work is focused on understanding the general scalability, measured by quality and time to solution, of these algorithms as we increase the size of the PCBO problems. If our research yields a hybrid quantum-classical algorithm competitive with or outperforming these approaches, even with a Hybrid D-Wave approach, it would likely be of intense interest to the broader optimization community.

3.2 D-Wave Quantum Annealers

D-Wave has set themselves apart by building devices with a large number of analog controlled qubits at the cost of not being able to easily generalize their systems to arbitrary computations. Its annealers are specialized for optimization, relying on the adiabatic theorem and low temperature thermalization effects to output heuristic solutions to a binary problem. The problems solved by the device take the form of spin variables, $\mathbf{s} \in \{0, 1\}^n$ which are encoded into an Ising model, which is an equivalent formulation to QUBO’s since a bijective mapping between the two forms can be made, $s_i = 1 - 2x_i$.

Since D-Wave has a constrained hardware graph, meaning not every qubit can interact with every other qubit, oftentimes a problem will have to be embedded using an embedding tool called *minorminer* which is able to heuristically find embeddings onto their hardware. Ideally these embeddings will be native, meaning that each variable in the input problem can be mapped to a single qubit; however it is often necessary to introduce logical chains. Logical chains are a series of qubits coupled with a strong ferromagnetic coupling, to produce a single effective qubit at long range. Unfortunately, since D-Wave cannot encode strict constraints, these chains function as penalty terms in the cost function, and therefore may be violated upon measurement. In this case, these "broken" chains will be corrected through classical post-processing to a feasible solution, often with a sub-optimal energy.

We tested the ability of both the D-Wave quantum annealer and hybrid solvers on an example 30-variable PCBO problem drawn from the **Pancan** data set. The results of these evaluations are shown in Figure 6. Note that our PCBO problems are all-to-all connected and therefore require general heuristic embeddings due to their size exceeding the maximum embedded complete graphs. Current implementations yield maximum chain lengths of 74 on Pegasus graphs [20], presenting challenges for solution quality. While D-Wave’s newly proposed Zephyr architecture [21] offers improved chain lengths of 54, hybrid solver approaches may provide more viable solutions. These are proprietary solvers and therefore we do not know exactly how they leverage quantum and classical computational resources. It is likely that they combine quantum sampling to warm-start classical refinement techniques in an effort to avoid local minima, but the specific quantum contribution remains unclear. Initial experiments with the *Leap_Hybrid_Sampler* solver demonstrate improved performance over direct QPU access, though not yet competitive with classical solvers like CPLEX.

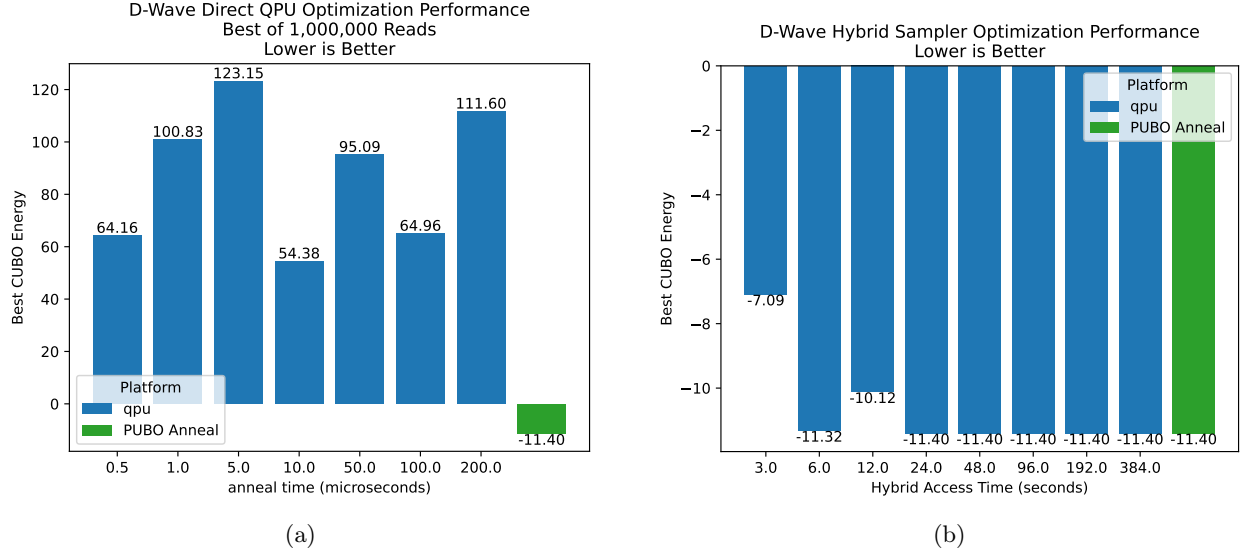


Figure 6: Results of PCBO evaluation on D-Wave (a) quantum annealers and the (b) *Leap_Hybrid_Sampler*. The PCBO is a 30-variable problem drawn from the **Pancan** data set and converted into a CUBO and then a QUBO via quadratization. “PUBO Anneal” is an implementation of simulated annealing available through the open source **qubovet** Python package [19], note that the X-axis time values do not apply to this platform.

However, D-Wave still has a few interesting options that we plan to explore throughout the rest of Phase 2. These include a Constrained Quadratic Solver, which would pass off handling and optimization of the solver constraints to D-Wave, allowing them to potentially use a classical solver that searches only in a feasible solution space, as well as a Nonlinear solver which would allow us to pass the Cubic problem directly to D-Wave. In each of these cases, since D-Wave’s hybrid solvers have more problem information to work with, they will likely yield higher quality solutions, though it should be again mentioned that it is difficult to tell how much of this performance is driven by the QPU rather than by classical solvers. We view our explorations in this area as a means to identify the current true state-of-the-art so as to inform our planned Phase 3 experiments aimed at demonstrating quantum advantage.

4 Resource Optimization

A significant challenge in implementing quantum algorithms for practical applications lies in the fundamental limitations of quantum computational hardware, particularly regarding gate operations. While quantum computers offer theoretical advantages for certain computational tasks, their practical utility is constrained by both slow clock speeds and imperfect gate operations [22, 23, 24, 25, 26]. These limitations impose practical caps on both the depth of circuits that can be reliably executed and the total number of circuits that can be run within reasonable timeframes. However, design choices at the application and compiler level can help to mitigate these hardware constraints.

In the first half of Phase 2 we addressed the primary bottleneck of our PCBO-Tournament algorithm: the iterative variational optimization loop appearing within the RQAOA. The degree to which we could mitigate this overhead would determine which quantum computing architectures could practically execute the PCBO-Tournament algorithm. Specifically, the large number of circuit evaluations incurred within the full variational loop would result in impractical runtimes for ion trap and neutral atom architectures which feature high operations with relatively slow gate speeds. Not addressing this challenge would leave the relatively fast superconducting architectures as the only viable option. However, the results of our research indicate that **the total number of quantum circuit executions required to run the PCBO-Tournament algorithm can be drastically reduced (e.g., by 1000X in Section 4.3.2) by leveraging a property**

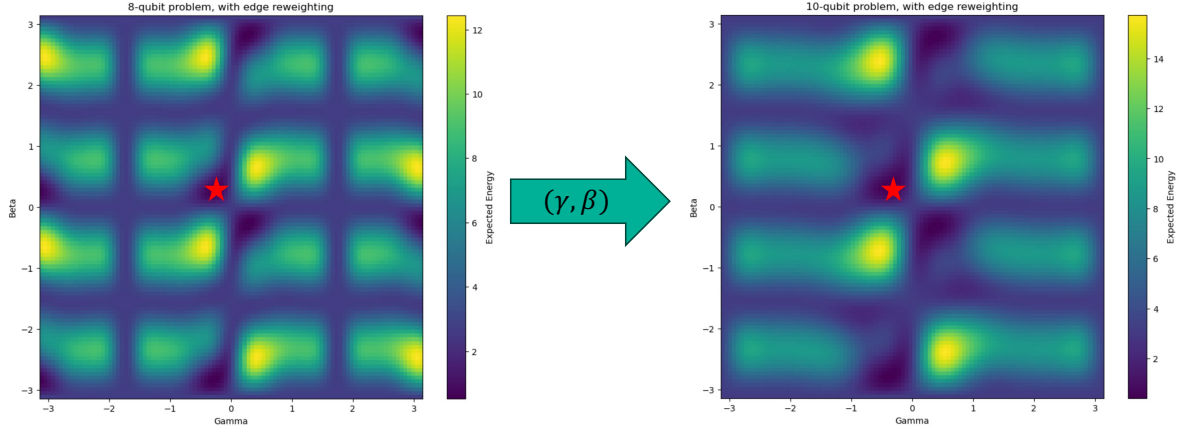


Figure 7: Illustrative example of parameter concentration and transfer between an (left) 8-qubit problem and a (right) 10-qubit problem with $p = 1$ depth for easy visualization. The two example problems are both drawn from the **Pancan** data set. Parameter concentration in the QAOA is the phenomenon that optimal variational parameters concentrate toward similar values across different problem instances of the same class. The star highlights a global minimum appearing near the same (γ, β) values in both problems. Therefore, it is possible to take the parameters found via expensive variational optimization on the smaller 8-qubit problem and *transfer* them to the 10-qubit problem – drastically reducing the quantum resource overhead required to obtain a solution on the 10-qubit problem.

known as *parameter concentration*. Reducing the overhead of the variational loop provides a clear path to scale up the size of our circuit simulations in the latter half of Phase 2 and to maintain broad options for future hardware experiments.

We collected our Phase 2 simulation data on Perlmutter [27] and the workstations acquired with Phase 2 funding. Each of these workstations contains an Intel Xeon w5-3435X CPU (45MB cache and 32 threads) and two 48GB Nvidia RTX 6000 Ada Generation GPUs. The target PCBO problems evaluated in Sections 4.2 and 4.3 were all drawn from the **Pancan** data set and constructed according to the T-PCBO formulation defined in Equation 2 with all coefficients $\lambda, \bar{\lambda}, \hat{\lambda}, \mu, \bar{\mu}, \hat{\mu} = 1$ set to one.

4.1 Parameter Concentration and Transfer

For hybrid variational algorithms [28], like the QAOA, the challenges of imperfect and slow operations are particularly apparent. Typically, variational algorithms require many circuit evaluations within an inner loop to measure the expectation value of the relevant cost function. This measurement is then used within an outer loop where a classical optimizer determines the parameter updates to begin the next iteration. For the PCBO-Tournament feature selection algorithm, we typically require fairly deep circuits, scaling with the order of the PCBO cost function, to generate high quality solutions. When these circuit execution requirements are combined with slow gate operations, the wall-clock runtime of these applications quickly becomes prohibitive.

Recent works address these limitations of the QAOA by leveraging a parameter transfer approach that strategically balances classical and quantum resources [29, 30, 31, 32, 33]. Specifically, the parameter transfer technique exploits a property of the QAOA whereby the value of the objective function for some fixed set of variational parameters $(\vec{\gamma}, \vec{\beta})$ will *concentrate* towards a single value across multiple different problems from the same class [29]. In recent years this phenomenon of parameter concentration has been studied theoretically and empirically mainly for MaxCut problems on random regular and general graphs [29, 30, 34, 31]. Importantly, parameter concentration implies that the optimal variational parameters for a small problem are likely to be very near to optimal for a larger problem, and therefore, the parameters may be *transferred* from the smaller problem to the larger one – effectively short circuiting the expensive variational loop on the large problem. In [30], for example, the authors demonstrated that the optimized parameters for a 6-node random graph could be successfully transferred to a 64-node problem instance with a less than

1% reduction in the approximation ratio. Figure 7 illustrates how parameter concentration allows for easy transfer between the energy landscapes of $p = 1$ QAOA for two example PCBO problems drawn from the **Pancan** data set.

The parameter transfer technique offloads the expensive variational optimization to classical computers, requiring only a small number of deep quantum circuits to be executed on the actual quantum hardware. This approach can be viewed as an expensive but valuable compilation technique, particularly justified for NP-hard optimization problems where solution quality warrants the classical overhead, like feature selection. By simulating many smaller tractable problems classically to extract optimal parameters, we can then execute a minimal number of deep circuits on the quantum computer to obtain solutions at the desired problem size.

Our work represents the first studies of parameter concentration in the RQAOA. Furthermore, while prior work has primarily considered MaxCut problems on random instances drawn from an abstract problem class, our work focuses on concrete PCBO problem instances constructed from real-world multimodal cancer data sets. In the following sections we present the results of our work studying the performance of parameter transfer techniques for the QAOA and the RQAOA applied to the third-order PCBO problems that appear within the PCBO-Tournament algorithm.

4.2 Parameter Transfer for the QAOA

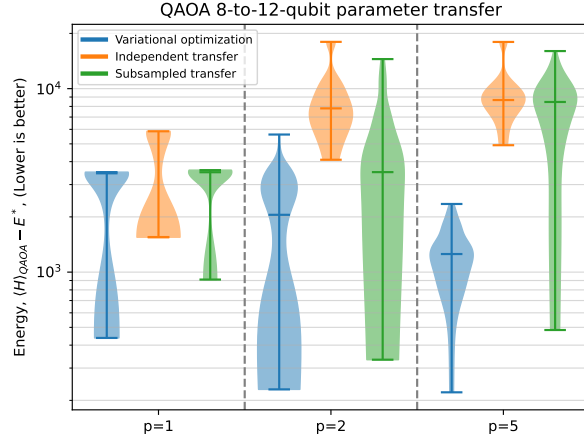
We evaluated the efficacy of parameter transfer for the QAOA by solving T-PCBO problems constructed over feature sets drawn from the **Pancan** data set. Ten subproblems were generated for each size by sampling features from the data set with a probability proportional to their mutual information with the target variable. Figure 8 contains the results of these evaluations. Our baseline evaluation was the full variational optimization (blue) of the QAOA circuits performed via the COBYLA optimizer implemented within the CUDA-Q library [35]. All variational optimizations were allowed a maximum of 5,000 iterations of the COBYLA optimizer.

We tested an “independent” parameter transfer (orange) technique where the variationally obtained $(\vec{\gamma}, \vec{\beta})$ parameters from a smaller problem were directly transferred to the larger target problem without any modification. We also evaluated a “subsamped” approach to parameter transfer (green) where a smaller problem was directly generated from the larger target problem by uniform random sampling over the variables in the larger problem. This subproblem contains all of the edges from the original PCBO problem whose constituent nodes also appear in the subproblem, and because the original PCBO is represented by a fully connected hypergraph (with first, second, and third-order edges) the sampled subproblem is also a fully connected hypergraph. Finally, the QAOA circuit was constructed and optimized over the subsampled problem and its final $(\vec{\gamma}, \vec{\beta})$ parameters were directly transferred to the larger target problem without any modification.

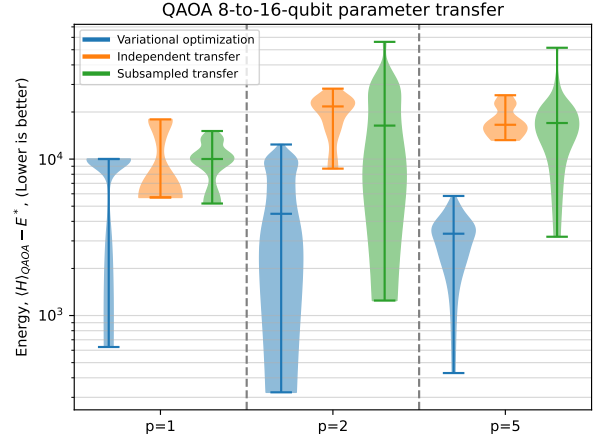
In Figure 8 we report the performance of these methods using the shifted energy $\langle H \rangle_{QAOA} - E^*$, where $\langle H \rangle_{QAOA} = \langle \psi(\vec{\gamma}, \vec{\beta}) | H | \psi(\vec{\gamma}, \vec{\beta}) \rangle$ is the expectation value of the QAOA circuit on the larger target problem using the $(\vec{\gamma}, \vec{\beta})$ parameters obtained by the specific method, and E^* is the energy of the globally optimal solution obtained via brute force search. We shift the energies in this way such that the best performance would have a value of zero on the plots. Unlike MaxCut, these PCBO problems do not admit a straightforward approximation ratio to report because their solutions may have negative, zero, or positive energies.

The $\langle H \rangle_{QAOA}$ expectation values have a wide range of values because they represent a weighted average over all possible 2^N solutions to the problem. Recall that QAOA prepares the state $|\psi(\vec{\gamma}, \vec{\beta})\rangle = U^{-i\beta_p} X U^{-i\gamma_p} H \dots U^{-i\beta_1} X U^{-i\gamma_1} H |+\rangle$. This state, if the variational optimization was performed successfully, will place higher probability on “good” low energy solutions and lower probability on “bad” high energy solutions. However, the variational optimization is approximate and therefore the resulting $|\psi(\vec{\gamma}, \vec{\beta})\rangle$ state will tend to have a nontrivial weight on poor solutions which will drive up the value of the $\langle H \rangle_{QAOA}$ expectation value. In practice, a single solution may be obtained by sampling this final output state a fixed number of times and checking the value of each sampled solution in polynomial time and taking the one with lowest energy. The RQAOA, discussed in Section 4.3, actually makes this analysis much easier because the algorithm returns only a single bitstring solution at the end of its execution.

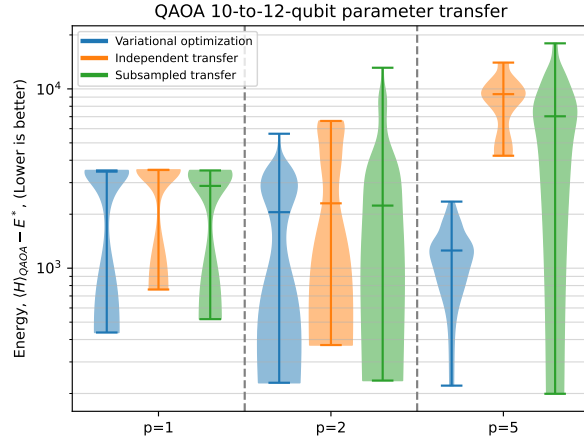
Figure 8 shows the evaluation results for optimizing 12- and 16-qubit QAOA circuits and transferring



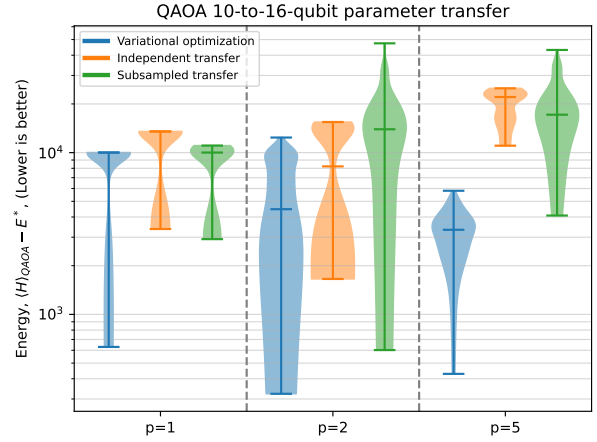
(a) Number of trials for 12-qubit variational optimization (blue): 191 at $p = 1$, 197 at $p = 2$, 94 at $p = 5$.



(b) Number of trials for 16-qubit variational optimization (blue): 260 at $p = 1$, 200 at $p = 2$, 101 at $p = 5$.



(c) Number of trials for 12-qubit variational optimization (blue): 191 at $p = 1$, 197 at $p = 2$, 94 at $p = 5$.



(d) Number of trials for 16-qubit variational optimization (blue): 260 at $p = 1$, 200 at $p = 2$, 101 at $p = 5$.

Figure 8: Violin plots showing the distribution of results obtained for QAOA on **Pancan** subproblems – the horizontal bars denote maximum, median, minimum values. Every Independent (orange) and Subsampled (green) transfer violin plot contains 100 distinct trials. The number of trials for each variational optimization (blue) violin plot is listed below each plot. The number of trials varied due to differing computational overheads and available compute time. Overall we observe that the full variational optimization performs the best and subsampled transfer tends to outperform independent transfer.

parameters to these problems from smaller 8- and 10-qubit circuits. Direct variational optimization (blue) achieves the lowest overall expectation values as well as consistently lower median values, which is the expected behavior because the parameters were trained specifically for those problems. Comparing the two parameter transfer techniques, the subsampled approach consistently yields lower expectation values and often has a lower median value than the independent transfer method. However, the subsampled parameter transfer results also have higher variance which we suspect may be caused by the uniform sampling over the target problem’s variables. It is likely that, for each target problem, there are certain variables which are more or less important – possibly correlated with the average weight of the edges that each node participates in – and, therefore, an important next step for future work is to develop more refined and effective heuristics for subproblem generation.

Future work may also explore the effect of fine-tuning of the larger QAOA ansatz after the parameters have been transferred, we include the results of a small fine-tuning experiment in Section 4.3.1. If parameter

transfer can effectively initialize the larger ansatz within a good local minima, then it may be feasible to greatly improve the expectation value of the ansatz with only a few iterations of gradient descent to travel to the bottom of that local minima. Interestingly, in the RQAOA parameter transfer results discussed in the next section, we see that RQAOA may be more forgiving of the quality of the parameter transfer in any particular round – i.e., we find many instances where transferring QAOA parameters in each recursive step of the RQAOA results in a higher quality solution than was obtained by direct variational optimization of the target problem.

4.3 Parameter Transfer for RQAOA

In the previous section we saw that transferring parameters for a single instance of QAOA can be effective, but typically results in suboptimal expectation values with a larger variance in performance. In this section we describe two approaches to transferring parameters for the RQAOA. In the RQAOA, parameter transfer is more nuanced than the QAOA because the technique must contend with the possibility that the “ensemble” in which the parameters concentrate is changing at each recursive iteration.

4.3.1 Lateral-RQAOA

Perhaps the most straightforward way to apply parameter transfer to the RQAOA is what we refer to as “Lateral-RQAOA”. In this approach, we assume that we have already spent the computational resources necessary to solve a single PCBO problem over N variables via the usual variational optimization of RQAOA. With this run saved, we can “laterally” transfer the optimized parameters from each recursive step to the same step of another problem of size N . This approach is similar in spirit to the independent parameter transfer for the QAOA discussed in Section 4.2 because the donated parameters are coming from another subproblem *independently* generated from the same data set (in our case **Pancan**). The reason why we cannot laterally transfer parameters between problems of different sizes is because the RQAOA will have fewer total recursive steps for the smaller problem (assuming the same cutoff value). Although Lateral RQAOA provides an incredible speedup by completely eliminating any variational optimization, this speedup can only be utilized *after* adequately solving a problem of the same difficulty.

For completeness we report the results Lateral-RQAOA parameter transfer in Figure 9 which shows a heatmap denoting the relative improvement of Lateral-RQAOA compared to the baseline of full variational optimization. Each subproblem index denotes a different N -variable problem and the parameters at each recursive step of RQAOA are donated from the “donor subproblem” to the “receiver subproblem”. Note that the main diagonal is always zero because it represents transferring parameters from a problem to itself. The final color of each cell represents a comparison between the solution obtained by laterally transferring parameters E_{don} and the *best* solution E_{opt} obtained by the COBYLA optimizer over an ensemble independent trials from randomly initialized starting parameters. The difference between the solution obtained via parameter transfer (E_{don}) and via variational optimization (E_{opt}) is scaled by the absolute value of the global minimum energy E^* found via brute force search. The maximum value in the heatmaps is clipped at a value of $\frac{E_{don}-E_{opt}}{|E^*|} = 2$ in order to better resolve the behavior of Lateral-RQAOA when it is competitive with the variationally optimized solution. We found that the color bar can be greatly skewed when parameter transfer does not yield a competitive solution (i.e., $\frac{E_{don}-E_{opt}}{|E^*|} > 1,000$). Conversely, we observed that Lateral-RQAOA can also produce very competitive solutions, outperforming the best variationally optimized solution over an ensemble of trials – up to a 17.5X improvement in energy for the ($donor = 6, receiver = 7$) instance in Figure 9f.

We also used Lateral-RQAOA to study the impact of “fine-tuning” on parameter transfer for RQAOA. In Figure 10 we show the same parameter transfer experiments as in Figure 9, but we allow for an extra fine-tuning step that updates the transferred parameters for a small number of additional optimizer iterations. Specifically, given the set $(\vec{\gamma}_{tr}, \vec{\beta}_{tr})$ of transferred parameters, we ran a small (characterized as a maximum cap of 100 optimizer iterations) additional optimization step on four free parameters (in contrast to the usual $2p$ seen in a full optimization) resulting in a new set of fine-tuned angles, $(\vec{\gamma}_{ft}, \vec{\beta}_{ft})$, defined as:

$$\vec{\gamma}_{ft} = \eta_0 \vec{\gamma}_{tr} + \delta_0 \vec{1}_p \quad (17)$$

$$\vec{\beta}_{ft} = \eta_1 \vec{\beta}_{tr} + \delta_1 \vec{1}_p \quad (18)$$

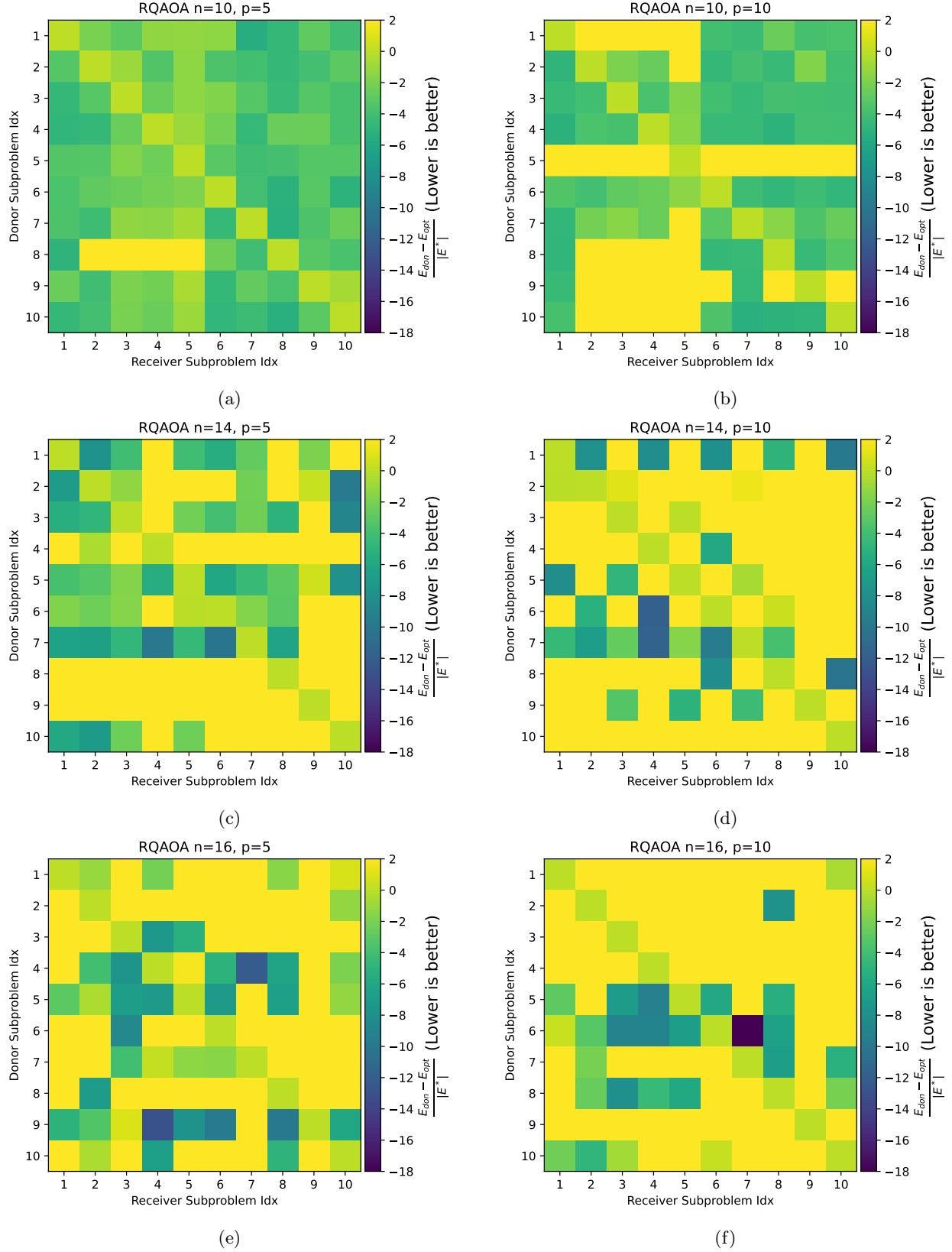


Figure 9: Heatmaps of Lateral-RQAOA (with no modifications to transferred parameters) compared to baseline RQAOA. E^* , E_{don} , E_{opt} are, respectively, the energies of the globally optimal solution, the solution obtained by Lateral-RQAOA, and the solution obtained by the best baseline RQAOA trial.

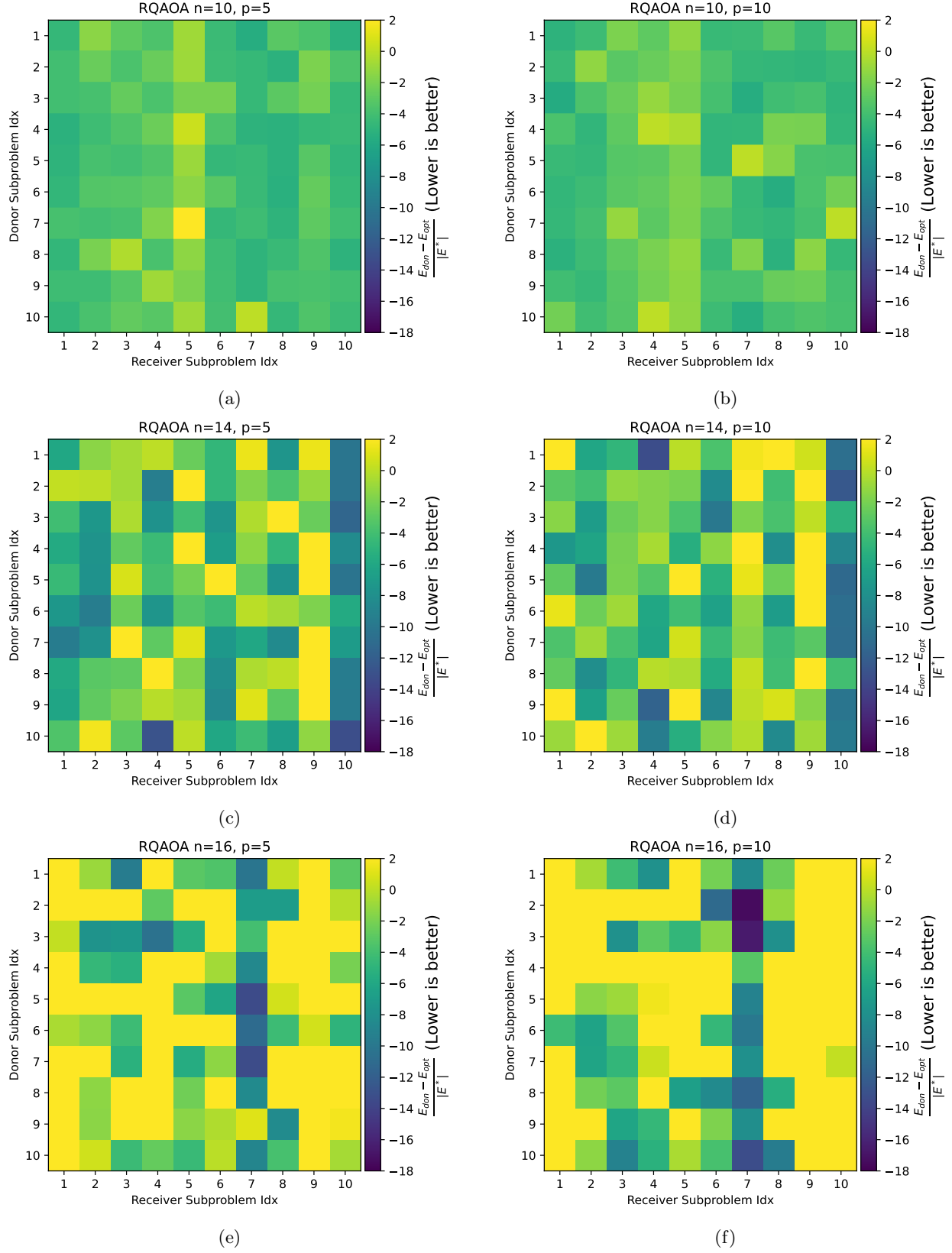


Figure 10: Heatmaps of Lateral-RQAOA (after the angle scale and offset optimized, or ‘fine-tuned’ with 100 maximum optimizer iterations) compared to baseline RQAOA. E^* , E_{don} , E_{opt} are defined the same as in Figure 9.

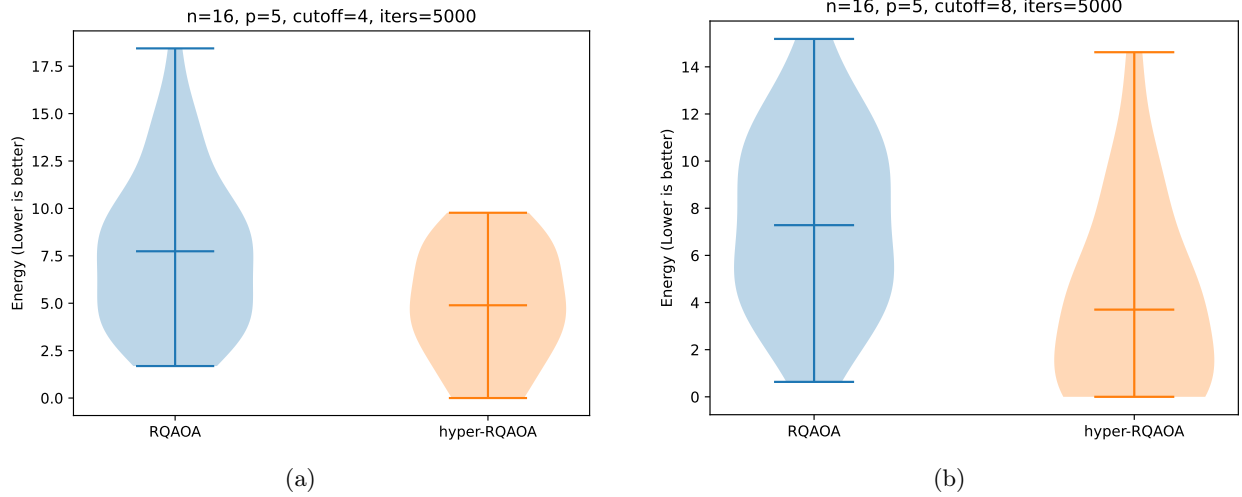


Figure 11: Performance comparison between RQAOA and hyper-RQAOA on 16-variable problems drawn from the **Pancan** data set, with depth $p = 5$, and a maximum of 5,000 iterations of the COBYLA optimizer for each variational loop. Violin plots denote the maximum, median, and minimum values of each ensemble. Hyper-RQAOA achieves comparable or lower energy solutions using a fraction of the computational resources compared to RQAOA. In each recursive step, Hyper-RQAOA generates and optimizes 5 subproblems containing a number of qubits equal to the cutoff and only simulates 5 instances of the larger circuit. In contrast, in each recursive step, RQAOA simulates up to 5,000 instances of the larger circuit – constituting a **1000X reduction in resource overhead per recursive step**.

where $\eta_i, \delta_i \in \mathbb{R}$ for $i \in \{0, 1\}$ and $\vec{1}_p$ is $(1, 1, \dots, 1)^T$ of length p . Note that with the fine-tuning of the scale (η_i) and offset (δ_i), the optimization bounds were removed; that is, there may be $[\gamma_{\text{ft}}]_j, [\beta_{\text{ft}}]_j \notin [0, 2\pi]$ for $j \in \{0, 1, 2, \dots, p-1\}$. The definition of $(\vec{\gamma}_{\text{ft}}, \vec{\beta}_{\text{ft}})$ was motivated by the fact that if a direct transfer does not perform optimally, a simple one- or two-parameter rescaling may bridge the gap [34]. This heuristic is loosely guided by mean-field theory, where effective coupling strengths scale with system size [36]. Moreover, it may also help correctly scale the angles for different energy landscapes of the respective subproblem, with an offset helping to perturb out of any close local minima. There is also an additional possibility of using these transferred parameters (or even fine-tuned parameters) for gradient descent into greater minima if used as a warm-start approach.

4.3.2 Hyper-RQAOA

We addressed the limitations of Lateral-RQAOA by developing a method that enables parameters to be transferred from a smaller to a larger problem within each recursive step of RQAOA, drastically reducing the number of large quantum circuit simulations. We refer to this method as Hyper-RQAOA, and it is based on the same subsampling approach discussed in Section 4.2.

A pseudocode outline of Hyper-RQAOA is presented in Algorithm 1. The algorithm operates identically to the baseline RQAOA in its strategy of fixing edges until reaching a cutoff point. The difference is that instead of relying on direct variational optimization of an N -qubit circuit to obtain a final set of $(\vec{\gamma}, \vec{\beta})$ parameters, the Hyper-RQAOA optimizes multiple, smaller M -qubit instances in parallel and directly transfers the best parameters to the larger N -qubit circuit. Specifically, we set $\text{cutoff} = M$ to be the size of the largest quantum circuit we are willing to variationally optimize via classical simulation. Then, in each recursive step, starting from the initial N -qubit problem $\mathcal{P}$, we subsample a fixed number, K , of subproblems each containing M variables. The subsampling is carried out according to a function $\mathcal{S} \leftarrow \text{subsampler}(\mathcal{P}, M)$ that takes the current larger problem and the desired size M as input, and it outputs a smaller problem $\mathcal{S}$. For the results presented here we assume the $\text{subsampler}()$ function implements a uniform sampling over the problem variables, as was assumed in Section 4.2, but we expect that a more effective heuristic for generating the subproblems could be devised in future work.

Input: PCBO problem $\mathcal{P}$ over N variables, cutoff M , QAOA depth p , max_iters, number K of subproblems, and method `subsampler()` to produce subproblems

Output: A solution to $\mathcal{P}$ represented as a binary string of length N

```

fixed_edges  $\leftarrow$  NewList()
while  $N > M$  do
     $E^* \leftarrow$  null;  $\beta^* \leftarrow$  null;  $\gamma^* \leftarrow$  null /* initialize */
    /* Find good parameters by optimizing smaller QAOA circuits */
    for  $i \leftarrow 1$  to  $K$  do
         $\mathcal{S}_i \leftarrow$  subsampler( $\mathcal{P}, M$ )
        /* Optimize variational parameters for the smaller problem */
         $\beta, \gamma \leftarrow$  optimize_QAOA_circuit( $\mathcal{S}_i, p, \text{max\_iters}$ )
        /* Calculate the energy of the parameters on the current PCBO */
         $E \leftarrow$  measure_expectation_on_ $H_{\mathcal{P}}$ ( $\gamma, \beta, \mathcal{P}, p$ )
        if ( $E^*$  is null) or ( $E < E^*$ ) then
            |  $\beta^* \leftarrow \beta$ ;  $\gamma^* \leftarrow \gamma$ ;  $E^* \leftarrow E$  /* update */
        end
    end
    /* Evaluate edge to fix using the best parameters trained on smaller problems */
    best_edge  $\leftarrow$  get_best_edge( $\mathcal{P}, \beta^*, \gamma^*$ )
    AddItem(fixed_edges, best_edge)
    /* Reduce problem size by one variable by fixing the best edge */
     $\mathcal{P} \leftarrow$  fix_best_edge( $\mathcal{P}, \text{best\_edge}$ )
     $N \leftarrow N - 1$ 
end
/* Optimally solve reduced problem */
partial_solution  $\leftarrow$  solve_bruteforce( $\mathcal{P}$ )
/* Substitute variables with fixed edges to reconstruct full solution */
solution  $\leftarrow$  reconstruct_full_solution(partial_solution, fixed_edges)
return solution

```

Algorithm 1: Hyper-RQAOA

Each of the subproblems, $\mathcal{S}_i \in \{\mathcal{S}_1, \dots, \mathcal{S}_K\}$, are classically optimized in parallel, and their final $(\vec{\gamma}_i, \vec{\beta}_i)$ parameters are transferred back to the larger problem. For each set of transferred parameters, $(\vec{\gamma}_i, \vec{\beta}_i) \in \{(\vec{\gamma}_1, \vec{\beta}_1), \dots, (\vec{\gamma}_K, \vec{\beta}_K)\}$, we compute $\langle \psi(\vec{\gamma}_i, \vec{\beta}_i) | H_{\mathcal{P}} | \psi(\vec{\gamma}_i, \vec{\beta}_i) \rangle$ on the quantum computer (or via classical simulation for the results shown here), where $H_{\mathcal{P}}$ is the Hamiltonian corresponding to the current larger problem $\mathcal{P}$.

The set of transferred parameters yielding the lowest expectation value $(\vec{\gamma}_{best}, \vec{\beta}_{best}) = \arg \min_{(\vec{\gamma}_i, \vec{\beta}_i)} \langle H_{\mathcal{P}} \rangle$, are used to determine which edge in $\mathcal{P}$ to fix according to the usual RQAOA implementation [5]. Once that edge is fixed, the problem size is reduced by one variable and we repeat these steps again until we reach the cutoff $= M$ and the problem is finished via brute force search. Finally, with the solution of the M -variable problem, the final solution to the N -variable problem is obtained by working back through each recursive step and making substitutions according the edges that were fixed.

Figure 11 shows a comparison between RQAOA and Hyper-RQAOA over an ensemble of $N = 16$ variable problems. Notably, the variational loops within both algorithms are given 5,000 iterations – this is 5,000 iterations of the COBYLA optimizer for each recursive step. Importantly, the RQAOA is optimizing quantum circuits containing 16, 15, 14, ... qubits all the way down to the cutoff. In these simulations, the Hyper-RQAOA is always generating and variationally optimizing $K = 5$ subproblems that each contain a number of qubits equal to the cutoff. It is transferring the parameters from these optimizations to initialize larger quantum circuits containing 16, 15, 14, ... qubits. Furthermore, despite the fact that the RQAOA is consuming a much greater amount of computational resources, we observe that the Hyper-RQAOA still finds solutions with lower energy. To illustrate the resource savings afforded by Hyper-RQAOA: in just the first recursive step on these example problems, RQAOA is evaluating 5,000 distinct 16-qubit circuits and Hyper-RQAOA is evaluating five distinct 16-qubit circuits – a 1000X reduction in resource overhead.

5 Understanding the Potential for Quantum Advantage on Cancer Data Sets

In tandem with our work developing the PCBO-Tournament feature selection algorithm we have also been collaborating with members of Professor Chong’s lab at UChicago to study the potential for quantum machine learning-based advantages on cancer data sets. This work is focused on understanding when and how a quantum computer might outperform classical models in the third step of the data processing pipeline from Figure 1.

Together with Professor Chong’s graduate students, we have shown that classical heuristics can be used to estimate the “strong k -contextuality” of empirical data sets. Strong k -contextuality is a novel quantity we introduce to quantify the difficulty of classically modeling a given data set. It is based on recent work proving that long-range correlations within a data set can lead to memory-based separations between quantum and classical machine learning models [37, 38]. These advantages stem from quantum contextuality – a property which arises from the fact that measurements *do not* simply reveal pre-existing properties of a system, but instead the observed properties depend on the choice of measurements made. Quantum contextuality is a fundamental property of quantum mechanics [39] and is also a crucial resource for quantum computation [40].

Strong k -contextuality is a property of a data set which places a lower bound on the memory size that a classical model must possess in order to accurately model the data. This means that a data set exhibiting strong k -contextuality contains patterns in its features which cannot be decomposed into fewer than k separate parts without losing essential relationships in the data. Strong k -contextuality is defined using the sheaf-theoretic formalism [41] to construct an empirical model composed of a set X of measurements and a set O of possible outcomes of those measurements. For example, the empirical model of a system of qubits could be described by the Pauli-X or Pauli-Z measurements we could make to each qubit and the $\{0, 1\}$ bit outcomes of those measurements. A collection of subsets of measurements $\mathcal{M}$ that form a cover of X are called a *measurement cover*, and the elements $C \in \mathcal{M}$ of the cover are sets of compatible measurements which can be performed simultaneously. We refer to each $C \in \mathcal{M}$ as a *measurement context*.

For any subset of measurements $U \subseteq X$, we have a function $s : U \mapsto O$ that assigns an outcome to each measurement — we call this a *section* over U . Let $\mathcal{E}(U)$ be the set of all possible sections over U , and

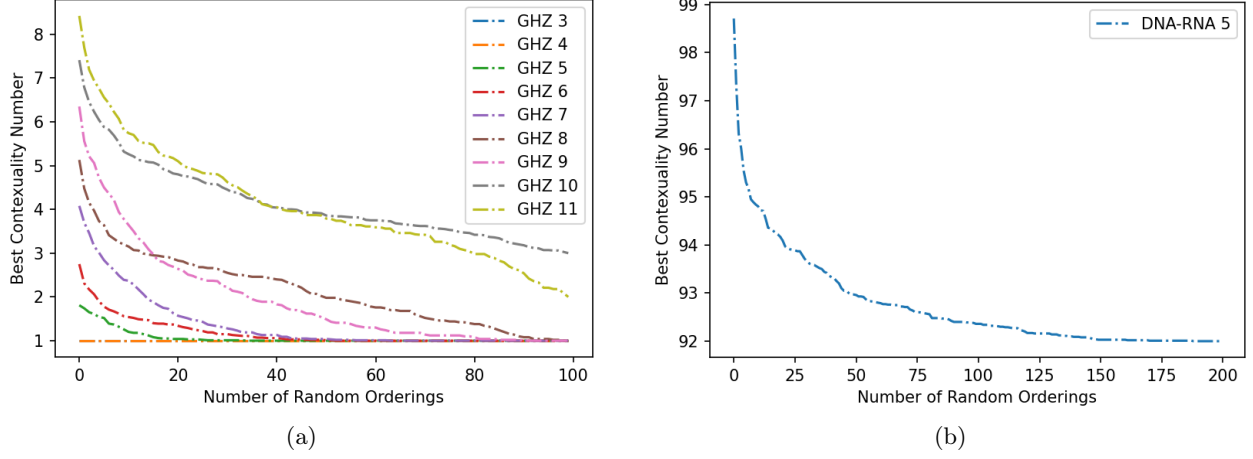


Figure 12: Performance of the greedy algorithm for computing contextuality numbers on (A) GHZ models and (b) a small DNA-mRNA model constructed from the **Pancan** data set.

further, let $\mathcal{D}_R\mathcal{E}(U)$ be the collection of all properly normalized probability distributions over the set $\mathcal{E}(U)$ of all possible sections.

An empirical model $\{e_C\}$ defines a probability distribution over the possible outcomes for each measurement within the context C . Concretely, for each context $C \in \mathcal{M}$, $e_C \in \mathcal{D}_R\mathcal{E}(C)$ defines a distribution of sections over the measurements in C .

The typical statement of contextuality refers to the ability to represent the empirical model using a *global section*, a section over all of X . Specifically, we say an empirical model is *strongly contextual* if the set

$$S_i := \{s \in \mathcal{E}(X) : \forall C \in \mathcal{M}, s_C \in \text{supp}(e_C)\}$$

is empty. In other words, a contextual empirical model does not admit a global assignment of outcomes to all the measurements. We extend this framework further by introducing *strong k -contextuality*. Instead of one global section, we allow for the empirical model to be distributed into k groups that cover $\mathcal{M}$. Let V_k (for $k \geq 1$) be the set of disjoint $\leq k$ -covers of $\mathcal{M}$; that is, each element of V_k is a set of cardinality at most k of disjoint $\mathcal{M}_i$ which cover $\mathcal{M}$. We define

$$S_e^k \equiv \bigcup_{\{\mathcal{M}_i\}_i \in V_k} \times_i S_e^{\mathcal{M}_i}, \quad (19)$$

i.e., the union over all disjoint k -covers $\{\mathcal{M}_i\}_i$ of $\mathcal{M}$ of Cartesian products of $S_e^{\mathcal{M}_i}$. Let k^* be the contextuality number—the largest k such that S_e^k is empty. We then say that e is *k^* -strongly contextual*.

In collaboration with the Chong Lab, we developed algorithms to approximate the contextuality number for cancer data sets. In general, computing this value is exponentially costly, and therefore our approach develops approximation algorithms based on greedy and graph coloring heuristics. We tested these algorithms on different empirical models where we were able to compute the true contextuality number via brute force enumeration and on an example DNA-mRNA data set where the true value is unknown. In practice, we found that both the greedy and the graph coloring algorithms gave similar results, but the runtime of the graph coloring heuristic has a strong dependence on the sparsity of the model which limits its broader applicability. For conciseness we only report the results of the greedy algorithm here.

Figure 12a shows the best contextuality number returned by the greedy algorithm as a function of the number of random orderings checked by the algorithm. If the algorithm were to check all exponentially many orderings then it would be guaranteed to find the true value. Therefore, this figure shows how quickly the greedy algorithm is able to converge to the true value as a function of the size of the empirical model. In Figure 12a, each color denotes a different GHZ N model which consists of a GHZ state $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\dots 0\rangle + |1\dots 1\rangle)$ over N qubits, the set of all combinations of Pauli-X and Pauli-Z measurements on the individual qubits, and the $\{0, 1\}$ outcomes of all those measurements. For the GHZ model, a global section can be found if the

empirical model is split into two groups that cover $\mathcal{M}$. Therefore, these models have a contextuality number of $k = 1$. For the smaller models, we see that the greedy algorithm does converge to this true value within 100 random orderings, but requires additional iterations for the 10- and 11-qubit models.

Finally, in Figure 12b we show the same greedy algorithm applied to an empirical model constructed from the **Pancan** data set. Specifically, the empirical model is constructed by taking the top five most correlated DNA-mRNA feature pairs from the **Pancan** data set. The set of measurements consists of all the binary strings of the DNA features (recall that these features denote the presence or absence of a particular mutation), and the observables correspond to the integer mRNA feature values (with values 0-4 denoting their relative expression level). At a high level, this empirical model represents an example learning problem where the task is to predict the expression level of the mRNA features using the string of binary DNA mutation features. In Figure 12b we see that the algorithm converges to a relatively high contextuality number. A classical model would require a memory space at least this large – i.e., at least $n = 7$ bits large such that $2^n \geq 92$ – to accurately model this learning problem.

While this particular example admits a relatively small classical model capable of solving the learning problem, we expect the contextuality number to grow rapidly as we expand the empirical model to encompass a larger number of multimodal cancer features. Interestingly, the lower bounds on the memory size of classical models obtained from the contextuality number do not apply to quantum machine learning models. Therefore, it would suffice to empirically demonstrate that a quantum model with a smaller memory is capable of solving the learning problem in order to achieve a quantum advantage. Our ongoing efforts are aimed precisely at this task – we are currently training classical and quantum models (specifically: hidden Markov models and basis-enhanced Bayesian quantum circuits [37]) on this learning task in order to empirically study their relative performance.

6 Ongoing Phase 2 Work, Future Phase 3 Plans, and Machine Targets

Thus far, our Phase 2 research has resulted in a first of its kind demonstration of effective parameter transfer through Hyper-RQAOA – yielding lower energy solutions and 1000X resource reduction compared to the baseline RQAOA (Figure 11) – as well as 25% improvement in F1 score for a 30-cancer tissue of origin classification task when selecting 6-feature sets over the whole **Pancan** data set (Figure 2) and 7% improvement when selecting 7-feature sets on a 30-feature **Pancan** subproblem (Figure 3). These results, and the relative difficulty of these optimization problems for current classical solvers, give us confidence that we can demonstrate quantum advantage in Phase 3 by (1) leveraging parameter transfer to evaluate a modest number of quantum circuits with more than 50 qubits and (2) relying on error mitigation, detection, and correction techniques to boost fidelity.

Parameter transfer techniques represent a significant advancement for scaling up evaluations of the RQAOA and PCBO-Tournament algorithm, both in simulation and hardware implementations. In the context of classical simulation, we were previously limited by the requirement of simulating thousands of circuits within the variational loop, which effectively limited experiments to around 20 qubits due to runtime constraints. Parameter transfer fundamentally lowers this resource overhead by allowing the majority of simulations to be performed on smaller circuits, with only a select few circuits requiring full-scale simulation. This shift in methodology pushes the practical limits to approximately 40–50 qubits, where GPU memory capacity, rather than runtime, becomes the primary limiting factor – representing a more efficient utilization of computational resources. In our ongoing Phase 2 work we plan to conduct simulation experiments leveraging parameter transfer to make comparisons with classical heuristic algorithms in this larger problem regime.

The benefits of parameter transfer extend beyond just simulation. They also enable larger-scale hardware evaluations across multiple quantum computing platforms. By shifting more of the computational workload to classical resources and reducing the number of required quantum circuits to mere tens rather than thousands, we can now target hardware evaluations on quantum computing architectures with relatively high gate fidelities but slow gate speeds, such as ion traps and neutral atoms, in addition to architectures with especially fast gate speeds such as superconducting quantum computers. This approach is particularly powerful when combined with platform-specific optimizations, such as those implemented within Infleqtion’s Superstaq com-

piler [2]. In cases where circuit depth is more constrained than qubit count, mid-circuit measurements and teleportation can be leveraged to recompile the quantum circuits from their typical cubic-depth, linear-qubit configuration to achieve linear depth with cubic qubit count [42, 43, 44]. This kind of circuit transformation is particularly well-suited to both IBM’s superconducting and Infleqtion’s SqaLe processors where qubit resources are relatively abundant compared to gate operations.

Finally, since parameter transfer allows us to focus on executing a smaller number of larger circuits, our future hardware evaluations of the PCBO-Tournament algorithm will leverage recent progress in the development and demonstration of error mitigation, detection, and correction in order to maximize the execution fidelity of these large quantum programs. For example, recent experiments from Riken and IBM [45] demonstrated state-of-the-art electronic structure calculations using a Heron processor combined with an array of error mitigation techniques [46]. Our applications are similarly amenable to such error mitigation strategies, especially given the Hamming weight constraint imposed when we desire to select N features. Progress has also been shown through demonstrations of error detecting codes used to improve the fidelity of multiple quantum architectures. Recent work [47] executed QAOA circuits on the Quantinuum H2-1 device [48] using the $[[k+2, k, 2]]$ “Iceberg” quantum error detecting code [49]. The experiments conducted in [47] demonstrated that the error detection capabilities of the Iceberg code yielded improvements to the encoded circuits compared to the unencoded version for circuits containing up to 20 logical qubits. More recently, Quantinuum also demonstrated 50 logical qubits on their current device [50]. Similarly, Infleqtion’s recent experimental demonstration on its SqaLe processor demonstrated ground state preparation using a $[[4, 2, 2]]$ Iceberg code realizing 12X reductions in error by encoding logical qubits into a larger number of physical qubits [1]. With Infleqtion’s multi-platform Superstaq software, we will target demonstrations on multiple modalities: our own neutral atom platform, Quantinuum, and IBM. It may even be possible to bootstrap parameters between platforms. Furthermore, we expect the quality of available quantum hardware platforms to steadily improve throughout the remainder of Phase 2 and a future Phase 3. IBM’s quantum computing roadmap includes the new Flamingo architecture to be released in 2025 containing 156 qubits on a single chip capable of executing 5,000 gates and with the potential to integrate multiple chips together into a 1,092-qubit modular architecture [51]. In 2025, Quantinuum’s roadmap includes the release of the new Helios architecture with 96 qubits with 2-qubit error rates below 5×10^{-4} , and capable of supporting approximately 50 logical qubits [52]. Finally, Infleqtion’s own roadmap includes the development of quantum computer capable of supporting approximately 30-100 logical qubits during the Phase 3 timeframe. We look forward to evaluating our cancer data feature selection application on these backends and pushing their performance to the limit through deep co-design of the compiler and hardware.

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